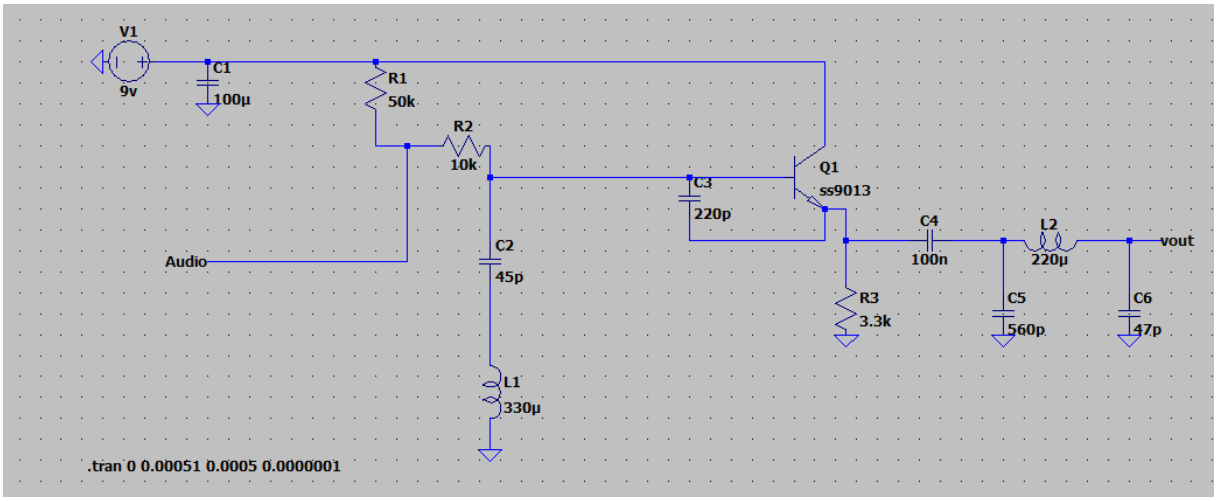
ELEC310 FALL 2024 Project

AM Transmitter Circuit

Yaşar Aykut Altuntop

Circuit Design

I based my design off this circuit I found on the internet:
https://leap.tardate.com/radio/am/singletransistortransmitter/



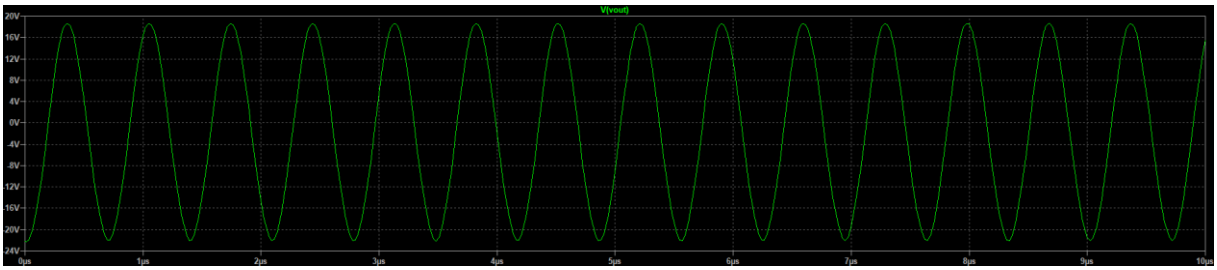
In the practical design, R1 is a potentiometer and C2 is a variable capacitor. C1 is used for a more stable DC current and to prevent in-rush current. C2 and L1 is used for carrier signal generation, for a wide range of frequencies C2 varies between 40-146pF. A single 20pF capacitor is connected to a 20-126pF variable capacitor. C5 is used as a coupling capacitor to allow the RF signal to pass and block the DC component.

The audio signal is introduced to the circuit before R2, joining with the carrier signal at the base of the transistor. After the transistor, R3 is used to allow the transistor to work in the active region and C4 is used as a coupling capacitor to pass the RF signals. The C5, L2, C6 part is a Pi filter used to attenuate noise to allow a cleaner signal.

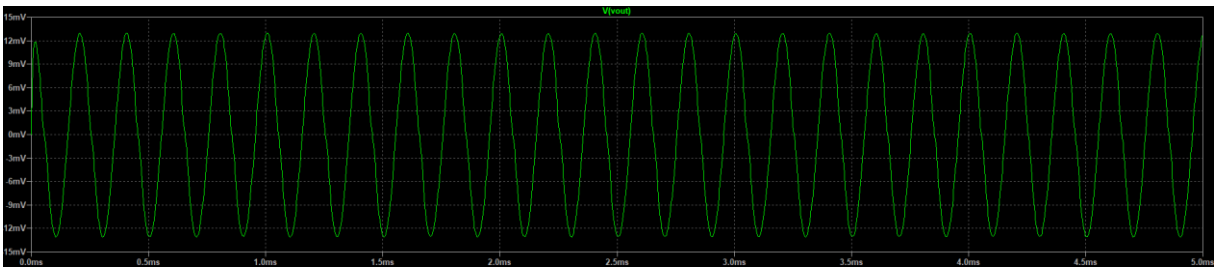
I used fixed biasing for the transistor, R2 is used to bias the base of the transistor, R3 ensures that the voltage at the collector is lower than the supply voltage and C3 enables the transistor to amplify the modulated signal without affecting its DC bias.

At max, the peak to peak voltage at the antenna was 20V making the the total gain 2.22.

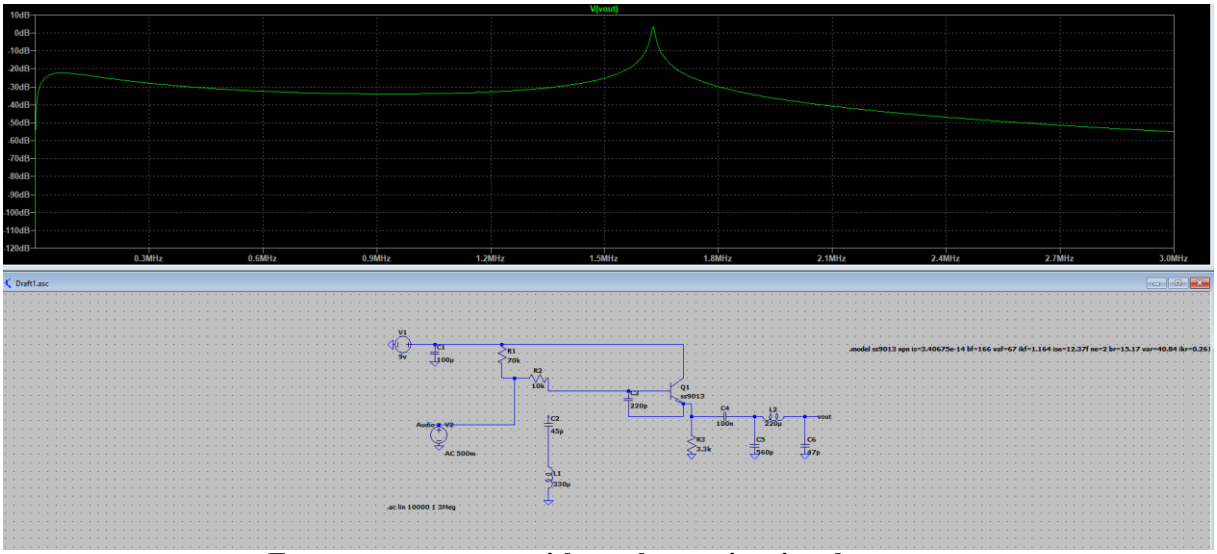
SPICE simulation of the circuit design



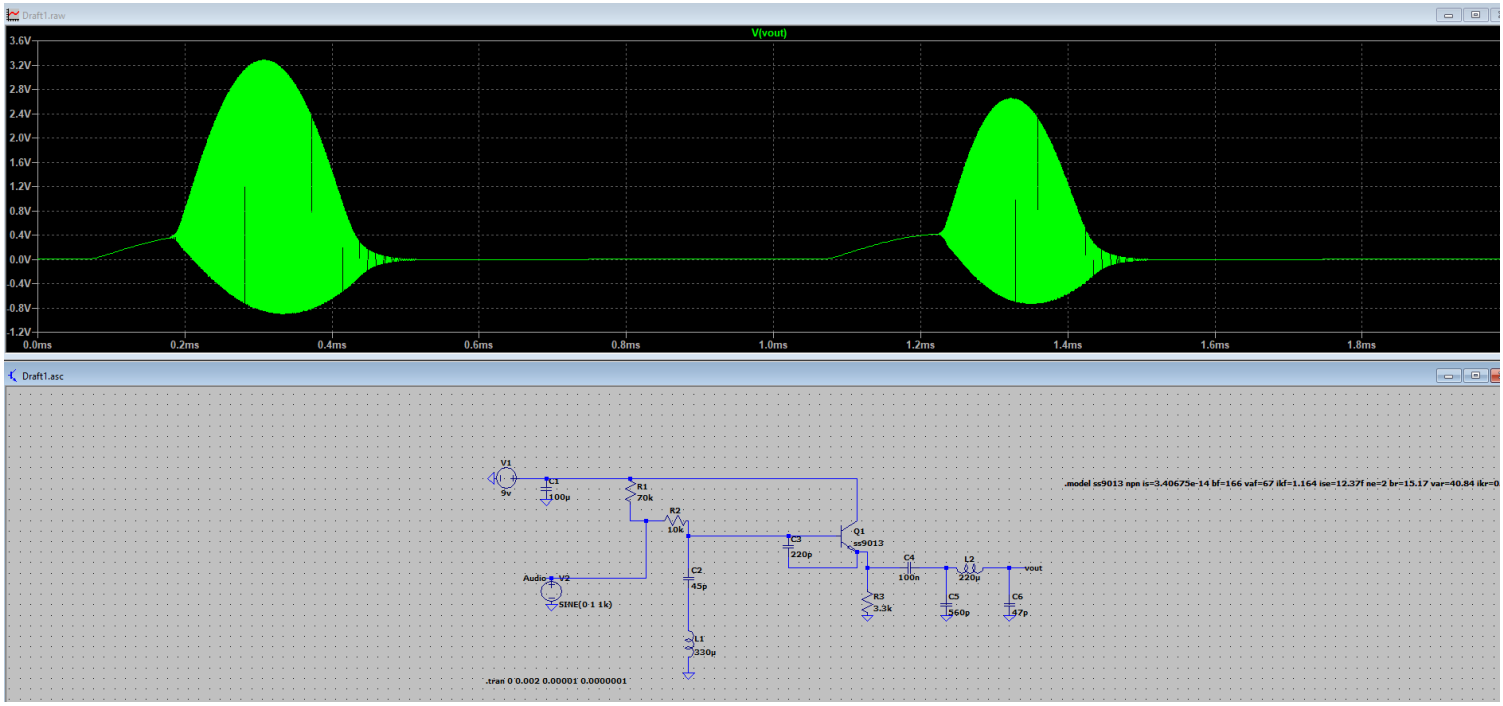
Voltage at the antenna with only carrier signal present, approximately around 1.3MHz



The voltage at the antenna with only 500mV 5kHz sine signal present

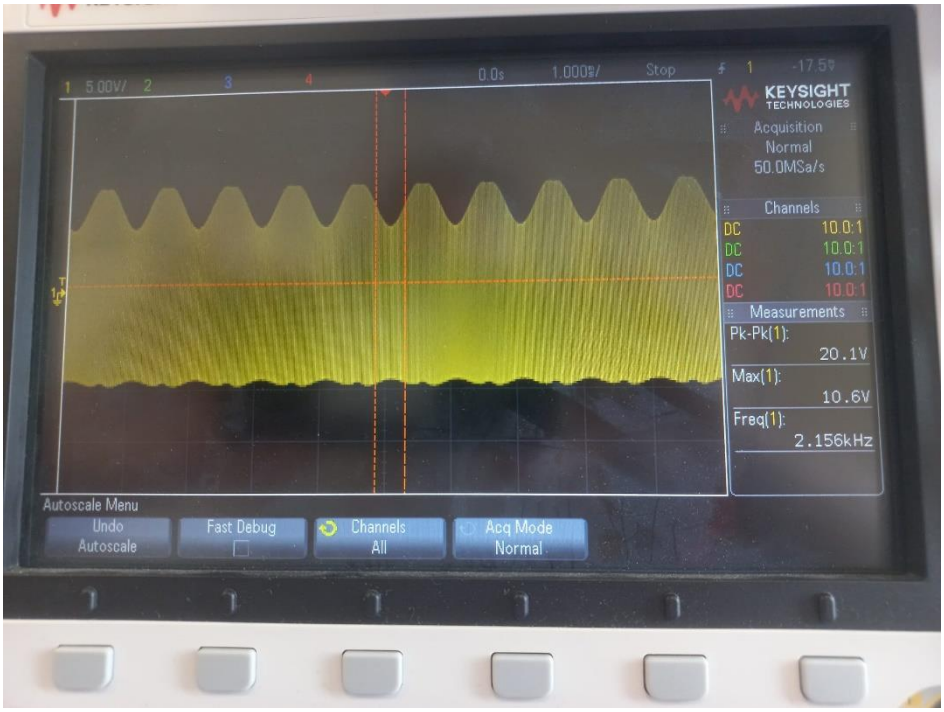


Frequency response without the carrier signal present.

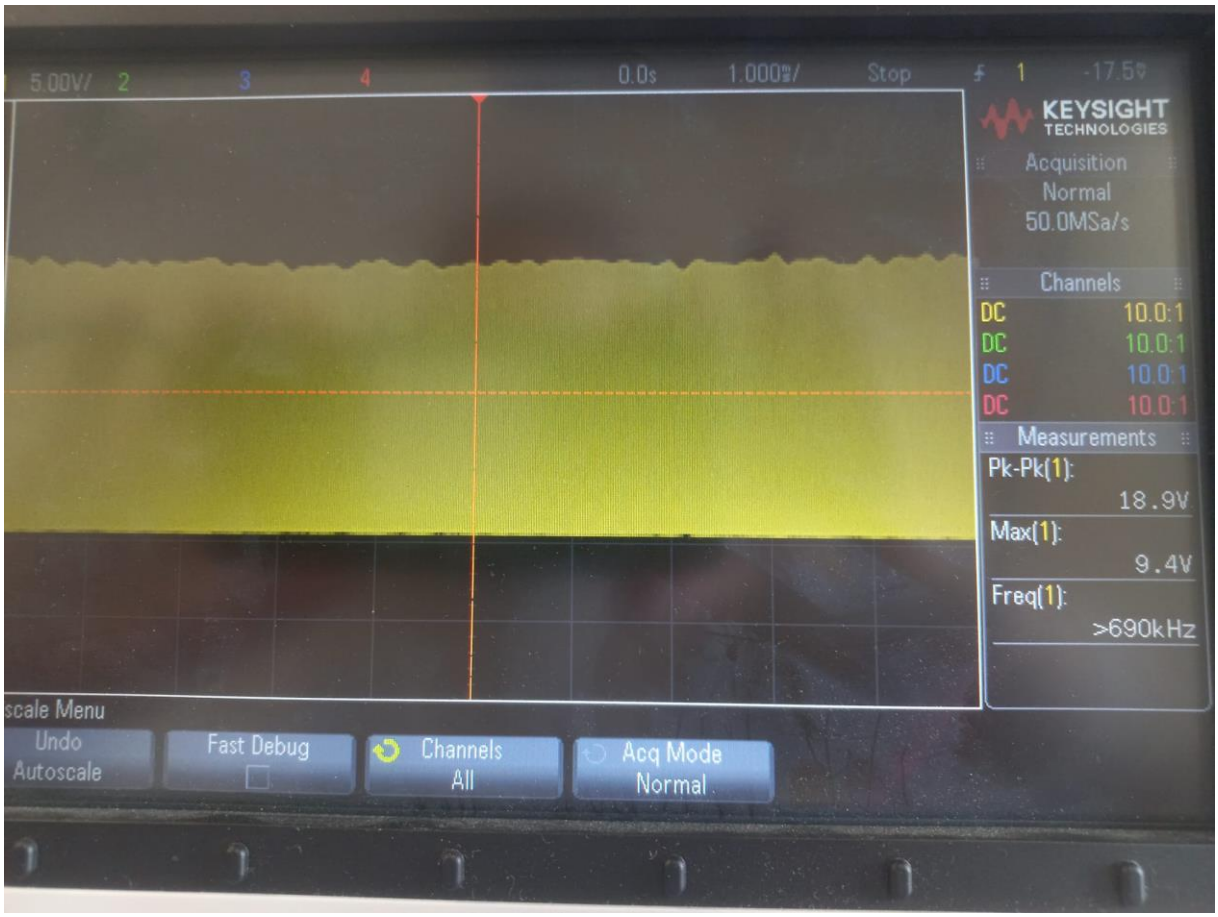


Plot of voltage generated at the antenna with both an input and a carrier signal present

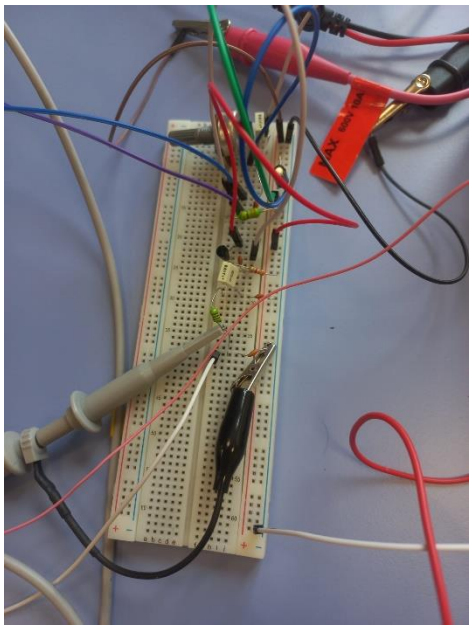
Measurements From Demo



Modulated 1kHz Sine Wave, the Sine Wave Amplitude was around 1V to show the modulation



Modulated Audio Wave



Circuit



Carrier Signal at 1.3MHz

Total gain was measured as 2.1 from the oscillator screen when audio is present.